

Philip P. Lampkin

philip.lampkin@utah.edu | <https://orcid.org/0000-0002-6908-9619> | <https://philip-lampkin.me>

Education

- Ph.D.** University of Wisconsin–Madison, Madison, WI June 2019 – Aug 2025
GPA: 4.0
Areas: Catalysis and Physical Organic Chemistry
- B.S.** University of San Francisco, San Francisco, CA Aug 2015 – May 2019
GPA: 3.98 (*Summa Cum Laude*)
Majors: Chemistry and Philosophy

Research

Postdoctoral Fellow, University of Utah, Department of Chemistry Jan 2026 – Present
Advisors: Matthew S. Sigman, Taylor D. Sparks, Jacob J. Lessard

- Discovered a new class of monomer (azabicyclobutanes) that embed azetidines into polymer backbones, enabling on-demand degradability or adaptability of materials using orthogonal stimuli.

Graduate Researcher, University of Wisconsin–Madison, Department of Chemistry June 2019 – Dec 2025
Advisor: Samuel H. Gellman

- Developed families of bifunctional organocatalysts with flexible and foldamer molecular scaffolds to assess the contributions of diad connection and preorganization to catalyst reactivity.
- Created an open-source platform for low-cost, high-performance 3D printed photoreactors.
- Led teams to recreate the Advanced Organic Chemistry Laboratory curriculum at UW–Madison.

Undergraduate Researcher, University of San Francisco, Department of Chemistry May 2017 – May 2019
Advisors: Claire Castro and William L. Karney

- Analyzed contributions of carbon atom tunneling to kinetics of isomerization reactions using DFT.
- Reported first instance of tunneling control where the preferred reaction is dominated by motions of heavy atoms.

Publications (Undergraduate Mentees, Graduate Mentees, (*) = Corresponding Author)

- Michel, C. S.; **Lampkin, P. P.**; Shezaf, J. Z.; Castro, C.*; Karney, W. L.* “Tunneling by 16 Carbons: Planar Bond Shifting in [16]Annulene” *J. Am. Chem. Soc.* **2019**, *141*, 13, 5286–5293. <https://doi.org/10.1021/jacs.8b13131>
- Girvin, Z. C.; **Lampkin, P. P.**; Liu, X.; Gellman, S. H.* “Catalytic Intramolecular Conjugate Additions of Aldehyde-Derived Enamines to α,β -Unsaturated Esters” *Org. Lett.* **2020**, *22*, 11, 4568–4573. <https://doi.org/10.1021/acs.orglett.0c01666>
- Lampkin, P. P.**; Thompson, B. J.; Gellman, S. H.* “Versatile Open-source Photoreactor Architecture for Photocatalysis Across the Visible Spectrum” *Org. Lett.* **2021**, *23*, 13, 5277–5281. <https://doi.org/10.1021/acs.orglett.1c01910>
- Śmiłowicz, D.; Eisenberg, S.; Ahn, S. H.; Koller, A. J.; **Lampkin, P. P.**; Boros E.* “Radiometallation and photo-triggered release of ready-to-inject radiopharmaceuticals from the solid phase” *Chem. Sci.*, **2023**, *14*, 5038–5055. <https://doi.org/10.1039/D2SC06977F>
- Lampkin, P. P.***; Xu, A.; Esselman, B. J.; Schwarz, C. E.; Thompson, S. D.; Gellman, S. H. Hill, N.* “Energy Transfer Photocatalytic (*E*) → (*Z*) Isomerization of 2-Nitrocinnamaldehyde using an Inexpensive, 3-D Printed Photoreactor Monitored by ^1H NMR Spectroscopy and Computational Modeling” *J. Chem. Educ.*, **2024**, *101*, 9, 3949–3957. <https://doi.org/10.1021/acs.jchemed.4c00620> – **ACS Editors’ Choice Award**
- Lampkin, P. P.**; Gellman, S. H.* “Dual Activation Modes Enable Bifunctional Catalysis of Aldol Reactions by Flexible Dihydrazides” *J. Am. Chem. Soc.* **2025**, *147*, 4, 3731–3739. <https://doi.org/10.1021/jacs.4c16143>

7. **Lampkin, P. P.**; Sanders, K. M.; L. C. Garman; Guzei, I., A.; Gellman, S. H.* "Bifunctional Catalysis of Aldol Reactions by Dihydrazide Foldamers: Assessment of Conformational Preorganization" *J. Am. Chem. Soc.*, **2025**, *147*, 37, 33932–33943. <https://doi.org/10.1021/jacs.5c10914>
8. **Lampkin, P. P.**; **Roberts, R. C.**; **Czeslawski, B.**; Gellman, S. H.* "Bifunctional Catalysis of a Crossed Aldol Condensation by Diamines: Impact of Tether Composition and Length" *J. Org. Chem.* **2026**, *in press*.

Open-Source Projects

Wisconsin Photoreactor Platform (WPP)

Repository: <https://github.com/uw-madison-chem-shops/wisconsin-photoreactor/>

- An open-source architecture for low-cost, high-performance photoreactors for synthetic organic chemistry.
- WPP reactors are composed of a 3D-printed enclosure, an inexpensive LED light source, off-the-shelf electronics and a custom circuit board.
- Dimensions and features of WPP devices are readily varied, enabling users to produce photoreactors tailored to diverse experimental needs.
- Academic, industrial and instructional laboratories across the world use Wisconsin Photoreactors.

Teaching

Organic Chemistry Lecturer, University of Wisconsin–Madison Aug 2025 – Dec 2025

- Led 2 sections of Organic Chemistry I (CHEM 343), consisting of 6 x 1-hour lectures per week (3 per section) to ~500 undergraduates (250 per section).

Curriculum Development Volunteer, University of Wisconsin–Madison Jan 2022 – Dec 2025

- Developed, implemented and taught 3 new Advanced Organic Chemistry Laboratory instructional experiments focused on modern synthetic photocatalysis methods (4 semesters).
- A report on a photosensitized alkene isomerization instructional experiment was published in *J. Chem. Educ.* and received an *ACS Editors' Choice* award. I am one of two corresponding authors on this report.
- Two reports on instructional modules involving metallaphotoredox and organic photocatalysts are in preparation. Former mentees Riley Lehman and Bianca Czeslawski will be authors on these works.

Teaching Assistant, University of Wisconsin–Madison June 2019 – Jan 2022

- Led weekly Organic Chemistry Lab sections and pre-lab discussions (3 semesters).
- Taught biweekly discussion sections for Organic Chemistry I & II (3 semesters).

Awards, Honors and Funding

Postdoctoral

ACS Arthur C. Cope Postdoctoral Fellowship (National Award)	2025
One-U Responsible AI Initiative Postdoctoral Fellowship (University Award)	2025

Graduate

William B. Dickinson Fellowship in Organic Chemistry (Departmental Award)	2024
Dorothy Powelson Award for Advanced Achievement in Teaching (University Award)	2024
Graduate Student Faculty Liaison Committee – Travel Award (Peer award)	2024
Graduate Student Faculty Liaison Committee – Mentorship Award (Peer Award)	2023
Robert C. Doban Mentor Award (Departmental Award)	2022
Outstanding Chemistry Teaching Assistant Award (Departmental Award)	2022
NSF GRFP – Honorable Mention (National Award)	2020

Undergraduate

Dr. Mel Gorman Award in Chemistry (Departmental Award)	2019
The American Institute of Chemists Student Award (Departmental Award)	2019
Robert M. Makus Memorial Service Award in Philosophy (Departmental Award)	2019
Whitehead Summer Research Fellowship in Chemistry (University Award)	2018

Professional Service and Outreach

<i>TA Feedback Session: L&S Framework for Advancing Teaching Excellence</i> – PhD Student Representative UW–Madison Letters & Sciences - Teaching & Learning Administration https://teachlearn.ls.wisc.edu/ls-teaching-excellence/	Apr 2025
<i>TA Expectations Committee</i> – PhD Student Representative UW–Madison Chemistry Department	Mar - Sep 2025
<i>Demystifying Graduate School Dinner</i> – Event, Panelist UW-Madison Letters & Sciences - Honors Program https://honors.ls.wisc.edu/event/demystifying-graduate-school-dinner-interactive-qa/	Feb 2025
<i>Facilitating More Effective Teaching Teams</i> – Article, Contributor UW-Madison Letters & Sciences - Instructional Design Collaborative https://idc.ls.wisc.edu/ls-design-for-learning-series/more-effective-teaching-teams/	Dec 2024
<i>Better Teaching Through Collaborative Instructor-TA Partnerships</i> – Event, Panelist UW-Madison Teaching and Learning Symposium	May 2024
<i>The Benzine: A Graduate Student–Led Zine Celebrating Art and Chemistry</i> – Article, Coauthor ACS Graduate & Postdoc Chemist Magazine https://gpchemist.acs.org/lab-life/the-benzine.html	Sep 2021
<i>The Benzine</i> – PhD Student Organization, Founder and Head Editor UW–Madison Chemistry Department https://chem.wisc.edu/thebenzine/	Sep - Aug 2019 - 2022

Mentees

Graduates	Years mentored	Current Role
<i>Theodore Gierszel</i>	2023 – 2024	Chemistry Education PhD Student (Stowe Group, UW-Madison)
<i>R. Charles Roberts</i>	2024 – 2025	Organic Chemistry PhD student (Gellman Group, UW-Madison)
Undergraduates		
<i>Angie Xu</i>	2020 – 2022	Instructor (Teach for America)
<i>Riley F. Lehman</i>	2022 – 2023	Organic Chemistry PhD Student (University of California, San Diego)
<i>Bianca Czeslawski</i>	2023 – 2024	Organic Chemistry PhD Student (New York University)